QUALITY ASSURANCE & DEFECT MITIGATION DEPARTMENT

Annual Quality Inspection & Defect Rate Audit — FY2025

Executive Summary: Product defect rates and rework overhead were systematically minimized during FY2025 via the installation of Automated Optical Inspection (AOI) cameras and laser profilometers at every manufacturing checkpoint.

Quarter	Inspected Units	Surface Defects (%)	Dimensional Flaws (%)	Total Defect Rate (%)	Scrap Cost ($M)
Q1	242,500	5.2%	4.2%	9.4%	$1.82M
Q2	265,100	4.8%	3.9%	8.7%	$1.64M
Q3	308,400	2.7%	2.2%	4.9%	$0.88M
Q4	354,200	1.1%	1.0%	2.1%	$0.32M

Direct Correlation to Efficiency: The defect rate reduction from 8.7% in Q2 down to 2.1% in Q4 (a 6.6 percentage point decrease) eliminated line stoppage for rework, contributing directly to the 17.0 percentage point surge in production efficiency.